E-WASTE & ENVIRONMENTAL MANAGEMENT

Toxic Chemicals in E-Waste

Assignment EW-01 | Any Five

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Topic	Sources and effects of five hazardous elements in electronic waste

Prepared for academic study of hazardous substances in discarded electronics.

1. Introduction: Why e-waste needs careful handling

Electrical and electronic products contain metals, plastics, glass and chemical additives. When computers, phones, displays and other devices are discarded, they become electronic waste (e-waste). Safe recycling can recover useful materials, but uncontrolled dismantling, open burning and acid treatment can release hazardous substances into air, soil and water.

The Global E-waste Monitor reported that about 53.6 million tonnes of e-waste were generated worldwide in 2019, while only 17.4% was documented as formally collected and recycled [1]. This highlights the need for proper collection, authorised recycling and safer product design.

Typical exposure pathway

Stage	What may happen
Discarded device	A device may contain lead, arsenic, chromium compounds, cadmium or mercury.
Unsafe processing	Crushing, burning or acid leaching may release dust, fumes or dissolved metals.
Environmental movement	Contaminants can enter air, soil, groundwater, rivers and sediments.
Exposure	People may inhale dust or vapour, drink contaminated water or consume affected food.

Relevant regulatory controls

Instrument	Purpose
EU RoHS Directive 2011/65/EU	Restricts certain hazardous substances in electrical and electronic equipment.
India E-Waste (Management) Rules, 2022	Provides a framework for environmentally sound e-waste management and extended producer responsibility.
BIS IS 10500:2012	Provides drinking-water quality specifications, including limits for several metals.
Minamata Convention on Mercury	International agreement addressing mercury-related risks and emissions.

The next sections discuss lead, arsenic, chromium (VI), cadmium and mercury. Regulatory limits depend on the material, chemical form and applicable standard, so current official rules should be checked for compliance decisions.

2. Lead (Pb)

Lead is a heavy metal historically used in electronics because of its useful electrical and material properties. It remains in the environment and is particularly harmful to the developing nervous system.

Where it may be found in e-waste

• Tin-lead solder on older printed circuit boards (PCBs).
• Lead-containing glass in cathode-ray-tube (CRT) monitors and televisions.
• Lead-acid batteries in UPS and backup systems.
• Some cable coverings and component finishes.

Potential effects on human health

• May affect brain development, learning, attention and behaviour in children.
• Long-term exposure in adults is associated with kidney damage, high blood pressure and anaemia.
• Can cross the placenta and affect foetal development.
• Some absorbed lead remains stored in bones for a long time.

Potential effects on the environment

• Dust and fragments from CRT glass or circuit boards may contaminate soil.
• Poorly controlled burning can produce lead-containing fumes and dust.
• Lead does not biodegrade, so contaminated sites can remain hazardous.

Key point: People near informal recycling areas may be exposed when contaminated dust settles on soil or enters homes.

3. Arsenic (As)

Arsenic is a metalloid found in specialised electronic materials, particularly some semiconductor components.

Where it may be found in e-waste

• Gallium arsenide (GaAs) in selected LEDs, laser diodes and high-frequency chips.
• Some infrared detectors and specialised solar cells.
• Certain semiconductor manufacturing processes.
• Some older glass and display applications.

Potential effects on human health

• Long-term exposure to inorganic arsenic can cause skin changes and increase the risk of several cancers.
• Chronic exposure may affect cardiovascular, metabolic and nervous-system health.
• High acute exposure can cause severe gastrointestinal illness.
• Risk depends on the chemical form, dose and exposure route.

Potential effects on the environment

• Poorly managed chip crushing or chemical recovery may release arsenic-containing dust or wastewater.
• Arsenic can enter groundwater under certain soil and water conditions.
• Some crops can take up arsenic from contaminated environments.

Key point: Specialised chips should be sent to authorised e-waste handlers; informal acid treatment may expose workers and nearby communities.

4. Chromium (VI) [Cr(VI)]

Chromium is used in different chemical forms. Hexavalent chromium, Cr(VI), is of particular concern because some compounds are toxic and carcinogenic.

Where it may be found in e-waste

- Chromate-based anti-corrosion treatments on metal parts.
- Some plated or coated casings, screws and fasteners.
- Certain older magnetic recording media and pigments.
- Selected components with chromium-containing finishes.

Potential effects on human health

- Inhalation of some Cr(VI) compounds can cause lung cancer.
- Exposure may lead to skin irritation, allergic dermatitis and ulcers.
- The respiratory tract may be affected, causing irritation and breathing problems.
- High exposure may also harm organs such as the liver and kidneys.

Potential effects on the environment

- Cr(VI) can be more mobile in water than some other chromium forms.
- Improper disposal of coated parts may contaminate soil and groundwater.
- Chromium compounds can harm aquatic organisms.

Key point: Chemical form matters: Cr(VI) generally presents greater health concern than chromium(III), so hazardous coatings require appropriate handling.

5. Cadmium (Cd)

Cadmium is a toxic metal used in some rechargeable batteries and electronic components. It can remain in the body for a long time, especially after repeated exposure.

Where it may be found in e-waste

- Nickel-cadmium (Ni-Cd) rechargeable batteries.
- Some chip resistors, semiconductor devices and infrared detectors.
- Selected older display phosphors and plastic stabilisers.
- Some electrical contacts, switches and coatings.

Potential effects on human health

- Can damage kidneys and affect the body's management of minerals.
- Long-term exposure can weaken bones and increase fracture risk.
- Cadmium and certain cadmium compounds are classified as carcinogenic to humans.
- It leaves the body slowly, so repeated exposure can lead to accumulation.

Potential effects on the environment

- Cadmium may move more readily through acidic soils.
- Plants can take it up, allowing it to enter food chains.
- It is toxic to aquatic organisms and may persist at contaminated sites.

Key point: Cadmium-containing batteries should not be mixed with ordinary rubbish; separate collection lowers leakage and exposure risks.

6. Mercury (Hg)

Mercury can form vapour at room temperature. It was used in several lighting and switching applications, making careful handling of older devices important.

Where it may be found in e-waste

• Compact fluorescent lamps and fluorescent tubes.
• Cold-cathode fluorescent backlights in some older LCD displays.
• Certain older relays, switches and thermostats.
• Some button-cell batteries and specialised equipment.

Potential effects on human health

• Can harm the nervous system, memory, coordination and development.
• Exposure during pregnancy can affect the developing foetus.
• Some forms of mercury can damage the kidneys.
• Mercury vapour can be absorbed through the lungs.

Potential effects on the environment

• Released mercury can travel through the atmosphere and later settle on land or water.
• Microorganisms can convert some mercury into methylmercury, which builds up in aquatic food chains.
• Mercury can remain in sediments and continue to pose risks.

Key point: Broken fluorescent lamps and older displays should be collected separately and must not be crushed or burned.

7. Comparison of the five elements

Element	Typical source	Main health concern	Environmental concern
Lead (Pb)	Older solder, CRT glass, batteries	Neurodevelopmental effects; kidney and blood-pressure effects	Persistent soil and groundwater contamination
Arsenic (As)	GaAs chips, LEDs, lasers	Cancer risk and skin effects after chronic exposure	Groundwater movement and crop uptake
Chromium (VI)	Some corrosion-resistant coatings	Cancer risk from inhalation; dermatitis	Mobile compounds may affect water and aquatic life
Cadmium (Cd)	Ni-Cd batteries, selected components	Kidney and bone damage; cancer risk	Crop uptake and food-chain transfer
Mercury (Hg)	Fluorescent lamps, older LCD backlights	Nervous-system and developmental effects	Methylmercury accumulation in aquatic food chains

Common patterns and prevention

• Persistence: these elements do not biodegrade like many organic pollutants. They may change chemical form or move between environmental compartments, but contamination can last a long time.
• Unsafe processing: open burning, uncontrolled acid baths and manual crushing increase exposure risks.
• Vulnerable groups: informal recycling workers, children and pregnant people may face elevated exposure.
• Safer design: manufacturers should reduce hazardous substances where feasible and comply with applicable restrictions.
• Responsible collection: use authorised collection points and producer take-back systems instead of mixing electronics with household rubbish.

8. Conclusion

Lead, arsenic, hexavalent chromium, cadmium and mercury have been used in specific electronic components because of their technical properties. When equipment is discarded, poorly controlled handling can release these substances and harm people, wildlife and ecosystems. Safer product design, separate collection, producer responsibility and authorised recycling are important ways to reduce these hazards.

References

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